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SYSTEMS AND METHODS FOR MIMO RANK INDICATOR OVERRIDE

Abstract

A device may include a processor configured to receive a multiple-input and multiple-output (MIMO) rank indicator from a user equipment (UE) device. The processor may be further configured to determine that the received MIMO rank indicator is lower than an expected MIMO rank indicator; determine that the UE device satisfies one or more criteria for a MIMO rank indicator override; and override the MIMO rank indicator for the UE device, in response to determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator and determining that the UE device satisfies the one or more criteria for the MIMO rank indicator override.

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Background/Summary

BACKGROUND INFORMATION

[0001] To satisfy the needs and demands of users of mobile communication devices, providers of wireless communication services continue to improve and expand available services as well as networks used to deliver such services. One aspect of such improvements includes increasing data rates at which mobile communication devices are able receive and send information. Managing high data rates for wireless communication devices may pose various challenges.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 illustrates an environment according to an implementation described herein;

[0003] FIG. 2 illustrates exemplary components of a device that may be included in a component of an environment according to an implementation described herein;

[0004] FIG. 3 illustrates exemplary components of multiple-input and multiple-output (MIMO) rank indicator override system according to an implementation described herein;

[0005] FIG. 4 illustrates exemplary components of a user equipment (UE) device database according to an implementation described herein;

[0006] FIG. 5 illustrates a flowchart of a process for classifying a UE device as a candidate for MIMO rank indicator override according to an implementation described herein;

[0007] FIG. 6 illustrates a flowchart for performing a MIMO rank indicator override according to an implementation described herein; and

[0008] FIG. 7 illustrates an exemplary signal flow diagram according to an implementation described herein.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0009] The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements.

[0010] A cellular wireless network enables user equipment (UE) devices to connect to networks via a Radio Access Network (RAN) and a core network in order to communicate with other devices connected to the RAN, communicate with devices in other networks, access applications or services hosted by a provider network, and/or make use of other types of wireless communication services. For example, Fifth Generation (5G) core network may implement different types of services for UE devices, such as, for example, an enhanced Mobile Broadband (eMBB) service for Voice over Internet Protocol (VoIP) telephone calls and/or data sessions for accessing Internet websites; a massive Internet of Things (mIoT) service for Internet of Things (IoT) devices; an Ultra-Reliable Low Latency Communication (URLLC) service for mission critical devices such as medical monitoring devices, autonomous vehicles, industrial automation, etc.; and/or other types of wireless communication services. Each type of communication service may be associated with a different set of requirements.

[0011] An important feature of 5G networks is the use of multiple-input and multiple-output (MIMO) communication. MIMO communication includes sending and receiving multiple data signals simultaneously over a same radio frequency (RF) channel using multiple antennas and/or antenna elements. For example, a UE device and a base station in a RAN may each include an adaptive antenna array with multiple antenna elements. If different antenna elements of a transmitting antenna array are able to transmit data with different signal propagation properties (e.g., by having different polarizations, etc.), a receiving antenna array may be able to receive and decode multiple signals simultaneously, thereby increasing the throughput of the data being transmitted.

[0012] In an ideal scenario, transmitting antennas of an antenna array may be able to transmit simultaneous streams of data. However, in real world situations, interference between individual

signal paths may decrease the reliability of particular signal paths and reduce the total rate of transmission by the antenna array. In a communication channel with m transmitting antennas and n receiving antennas, the signal transmission may be modeled as an m by n channel matrix that includes all $m \times n$ paths between the transmitting antennas and the receiving antennas.

[0013] The rank of a matrix is a measure of the number of linearly independent vectors (i.e., rows or columns) of the matrix. Thus, the rank of a channel matrix may indicate how many data streams may be spatially multiplexed on a MIMO communication channel. The rank of a MIMO channel matrix may be referred to as a “MIMO rank.” As an example, a single antenna may be associated with a MIMO rank of 1. As another example, a 2×2 antenna array may be associated with MIMO rank values between a value of 2, corresponding to an ideal situation, and a value of 1, corresponding to a worst-case scenario in which the throughput of the 2×2 antenna array is no better than a single antenna. As yet another example, a 4×4 antenna array may be associated with MIMO rank values between 4 and 2, etc. Massive MIMO antenna arrays may have hundreds, or even thousands of antenna elements. Data sent via MIMO transmission may be mapped onto two codewords transmitted simultaneously. Even in MIMO transmission schemes with a MIMO rank higher than 2, two codewords may be used, as increasing the number of codewords beyond 2 may not significantly enhance performance.

[0014] A UE device may estimate the MIMO rank for a MIMO communication channel between the UE device and a base station by measuring reference signals transmitted by the base station. The UE device may determine a MIMO rank based on the measured reference signals and report the determined MIMO rank to the base station. The base station may then schedule downlink data transmission to the UE device based on the reported MIMO rank. For example, the base station may send downlink channel information (DCI) to the UE device in a Physical Downlink Control Channel (PDCCH) that enables the UE device to decode data transmitted to the UE device by the base station via a Physical Downlink Shared Channel (PDSCH). The DCI information may include the MIMO configuration selected by the base station to transmit the data.

[0015] A UE device may report a lower MIMO rank than the MIMO rank determined by the UE device based on measured reference signal. For example, a manufacturer of a wireless chipset for the UE device may intentionally configure the wireless chipset to report a lower MIMO rank in order to conserve battery power, since using fewer antenna elements requires using less power. However, using a lower MIMO rank than is available may result in inefficient use of network resources, because the base station may need to schedule more resource blocks to transmit the same amount of data to the UE device. Furthermore, other UE devices serviced by the base station may experience a lower throughput as a result.

[0016] Implementations described herein relate to systems and methods for MIMO rank indicator override. A device associated with a base station may be configured to override a MIMO rank indicator received from a UE device and transmit data to the UE device at a higher MIMO rank than the MIMO rank identified by the received MIMO rank indicator. For example, the device may be configured to receive a MIMO rank indicator from the UE device, determine that the received MIMO rank indicator is lower than an expected MIMO rank indicator, determine that the UE device satisfies one or more criteria for a MIMO rank indicator override, and override the MIMO rank indicator for the UE device, in response to determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator and determining that the UE device satisfies the one or more criteria for the MIMO rank indicator override.

[0017] Overriding the MIMO rank indicator for the UE device may include selecting a MIMO rank greater than the received MIMO rank indicator sent by the UE device and including information identifying the selected MIMO rank in DCI sent to the UE device. Receiving the MIMO rank indicator sent from the UE device may include retrieving the MIMO rank indicator from a measurement report sent from the UE device. Determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator may include determining the expected MIMO

rank indicator based on at least one of a Sounding Reference Signal (SRS) associated with the UE device or a Precoding Matrix Indicator (PMI) associated with the UE device.

[0018] Determining that the UE device satisfies one or more criteria for the MIMO rank indicator override may include determining that a channel quality value associated with the UE device is greater than a channel quality threshold, determining that a cell loading condition associated with the UE device is greater than a cell loading threshold, and/or determining that the UE device has been classified as a candidate for the MIMO rank indicator override. Determining that the UE device has been classified as the candidate for the MIMO rank indicator override may be based on at least one of an application in use by the UE device, a Quality of Service (QoS) parameter for a data flow associated with the UE device, a network slice in use by the UE device, a Multi-Access Edge Computing (MEC) service in use by the UE device, and/or a location associated with the UE device.

[0019] The device may be further configured to determine whether overriding the MIMO rank indicator was appropriate. Overriding the MIMO rank indicator may be appropriate if the higher MIMO rank does not result in an increased signal error rate as measured by the UE device. For example, the device may be configured to determine whether a block error rate (BLER) associated with the UE device is less than a BLER threshold. If the BLER is less than the BLER threshold, the base station may continue to override the MIMO rank indicator for the UE device, in response to determining that the BLER associated with the UE device is less than the BLER threshold. If the BLER is not less than the BLER threshold for at least a particular number of time slots or a particular time period, the base station may cease to override the MIMO rank indicator for the UE device and revert to a MIMO rank corresponding to the MIMO rank identified in the MIMO rank indicator received from the UE device. In some implementations, rather than reverting to a MIMO rank identified in the MIMO rank indicator, the base station may reduce the MIMO rank to a value between the overridden MIMO rank and the MIMO rank identified in the MIMO rank indicator.

[0020] FIG. 1 is a diagram of an exemplary environment **100** in which the systems and/or methods described herein may be implemented. As shown in FIG. 1, environment **100** may include UE devices **110-A** to **110-N** (referred to herein collectively as “UE devices **110**” and individually as “UE device **110**”), a RAN **130** that includes base stations **120-A** to **120-M** (referred to herein collectively as “base stations **120**” and individually as “base station **120**”), a Multi-Access Edge Computing (MEC) network **140**, a core network **150**, and packet data networks (PDNs) **160-A** to **160-Y** (referred to herein collectively as “PDNs **160**” and individually as “PDN **160**”).

[0021] UE device **110** may include any mobile device with cellular wireless communication functionality. UE device **110** may include a handheld wireless communication device (e.g., a mobile phone, a smart phone, a tablet device, etc.); a wearable computer device (e.g., a head-mounted display computer device, a wristwatch computer device, etc.); a laptop computer, a tablet computer, a portable gaming system, and/or another type of portable computer; a WiFi access point (AP); a Fixed Wireless Access (FWA) device; and/or any other type of mobile computer device with cellular wireless communication capabilities. In some implementations, UE device **110** may communicate using machine-to-machine (M2M) communication, such as Machine Type Communication (MTC), and/or another type of M2M communication for IoT applications.

[0022] Base station **120** may include a 5G New Radio (NR) base station (e.g., a gNodeB) and/or a Fourth Generation (4G) Long Term Evolution (LTE) base station (e.g., an eNodeB). Each base station **120** may include devices and/or components configured to enable cellular wireless communication with UE devices **110**. For example, base station **120** may include an RF transceiver configured to communicate with UE devices **110** using a 5G NR air interface using a 5G NR protocol stack, a 4G LTE air interface using a 4G LTE protocol stack, and/or using another type of cellular air interface.

[0023] RAN **130** may include base stations **120** and be managed by a provider of wireless communication services. RAN **130** may enable UE devices **110** to connect to core network **150** via

base stations **120** using cellular wireless signals. For example, RAN **130** may include one or more central units (CUs), distributed units (DUs), and/or Radio Units (RUs) (not shown in FIG. **1**) that enable and manage connections from RUs to core network **150**. RAN **130** may include features associated with an LTE Advanced (LTE-A) network and/or a 5G network or other advanced network, such as management of 5G NR base stations; carrier aggregation; advanced or massive MIMO configurations (e.g., an 8×8 antenna configuration, a 16×16 antenna configuration, a 256×256 antenna configuration, etc.); cooperative MIMO (CO-MIMO); relay stations; Heterogeneous Networks (HetNets) of overlapping small cells and macrocells; Self-Organizing Network (SON) functionality; MTC functionality, such as 1.4 Megahertz (MHz) wide enhanced MTC (eMTC) channels (also referred to as category Cat-M1), Low Power Wide Area (LPWA) technology such as Narrow Band (NB) IoT (NB-IoT) technology, and/or other types of MTC technology; and/or other types of LTE-A and/or 5G functionality.

[0024] MEC network **140** may be associated with one or more base stations **120** and may provide MEC services for UE devices **110** attached to the base stations **120**. MEC network **140** may be in proximity to base stations **120** from a geographic and network topology perspective, thus enabling low latency services to be provided to UE devices **110**. As an example, MEC network **140** may be located on the same site as base station **120**. As another example, MEC network **140** may be geographically closer to one of base stations **120** and reachable via fewer network hops and/or fewer switches, than other base stations **120**.

[0025] MEC network **140** may include one or more MEC devices **145**. MEC devices **145** may provide MEC services to UE devices **110**. A MEC service may include, for example, a low-latency microservice associated with a particular application, a microservice associated with a virtualized network function (VNF) of core network **150**, a cloud computing service, such as cache storage service, artificial intelligence (AI) accelerator service, machine learning service, an image processing service, a data compression service, a locally centralized gaming service, a Graphics Processing Units (GPUs) and/or other types of hardware accelerator service, and/or other types of cloud computing services.

[0026] Core network **150** may be managed by the provider of cellular wireless communication services and may manage communication sessions of subscribers connecting to core network **150** via RAN **130**. For example, core network **150** may establish an Internet Protocol (IP) connection between UE devices **110** and PDN **160**. In some implementations, core network **150** may include a 5G core network. The components of core network **150** may be implemented as dedicated hardware components and/or as Virtual Network Functions (VNFs) implemented on top of a common shared physical infrastructure using Software Defined Networking (SDN). For example, an SDN controller may implement one or more of the components of core network **150** using an adapter implementing a VNF virtual machine, a Cloud-Native Network Function (CNF) container, an event driven serverless architecture, and/or another type of SDN architecture. The common shared physical infrastructure may be implemented using one or more devices **200** described below with reference to FIG. **2** in a cloud computing center associated with core network **150**.

Additionally, or alternatively, some, or all, of the common shared physical infrastructure may be implemented using one or more devices **200** included in MEC network **140**.

[0027] Core network **150** may include a MIMO rank indicator override system **155**. MIMO rank indicator override system **155** may be configured to override a MIMO rank indicator received from UE device **110** by base station **120** and cause base station **120** to transmit data to UE device **110** at a higher MIMO rank than the MIMO rank identified by the received MIMO rank indicator. While MIMO rank indicator override system **155** is shown in core network **150** for illustrative purposes, in other implementations, MIMO rank indicator override system **155** may be implemented as part of base station **120**. In yet other implementations, MIMO rank indicator override system **155** may be implemented in a different device that is part of RAN **130**, such as a DU or a CU. In yet other implementations, MIMO rank indicator override system **155** may be implemented in MEC device

145.

[0028] PDNs **160-A** to **160-Y** may each be associated with a Data Network Name (DNN) in 5G, and/or an Access Point Name (APN) in 4G. UE device **110** may request a connection to PDN **160** using a DNN or an APN. For example, UE device **110** request a connection to an application server **165** (shown in PDN **160-A** for illustrative purposes). PDN **160** may include, and/or be connected to, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), an autonomous system (AS) on the Internet, an optical network, a cable television network, a satellite network, a wireless network, an ad hoc network, a telephone network (e.g., the Public Switched Telephone Network (PSTN) or a cellular network), an intranet, or a combination of networks. PDN **160** may include application server **165**. Application server **165** may include one or more computer devices that host one or more applications and/or other types of services used by UE device **110**. Core network **150** may establish a data flow session between UE device **110** and application server **165** via RAN **130**.

[0029] Although FIG. **1** shows exemplary components of environment **100**, in other implementations, environment **100** may include fewer components, different components, differently arranged components, or additional components than depicted in FIG. **1**. Additionally, or alternatively, one or more components of environment **100** may perform functions described as being performed by one or more other components of environment **100**.

[0030] FIG. **2** is a diagram illustrating example components of a device **200** according to an implementation described herein. Each of the components of FIG. **1** may include, or be implemented on, one or more devices **200**. As shown in FIG. **2**, device **200** may include a bus **210**, a processor **220**, a memory **230**, an input device **240**, an output device **250**, and a communication interface **260**.

[0031] Bus **210** may include a path that permits communication among the components of device **200**. Processor **220** may include any type of single-core processor, multi-core processor, microprocessor, latch-based processor, central processing unit (CPU), graphics processing unit (GPU), tensor processing unit (TPU), hardware accelerator, and/or processing logic (or families of processors, microprocessors, and/or processing logics) that interprets and executes instructions. In other embodiments, processor **220** may include an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or another type of integrated circuit or processing logic.

[0032] Memory **230** may include any type of dynamic storage device that may store information and/or instructions, for execution by processor **220**, and/or any type of non-volatile storage device that may store information for use by processor **220**. For example, memory **230** may include a random access memory (RAM) or another type of dynamic storage device, a read-only memory (ROM) device or another type of static storage device, a content addressable memory (CAM), a magnetic and/or optical recording memory device and its corresponding drive (e.g., a hard disk drive, optical drive, etc.), and/or a removable form of memory, such as a flash memory.

[0033] Input device **240** may allow an operator to input information into device **200**. Input device **240** may include, for example, a keyboard, a mouse, a pen, a microphone, a remote control, an audio capture device, an image and/or video capture device, a touch-screen display, and/or another type of input device. In some implementations, device **200** may be managed remotely and may not include input device **240**. In other words, device **200** may be “headless” and may not include a keyboard, for example.

[0034] Output device **250** may output information to an operator of device **200**. Output device **250** may include a display, a printer, a speaker, and/or another type of output device. For example, device **200** may include a display, which may include a liquid-crystal display (LCD) for displaying content to the user. In some implementations, device **200** may be managed remotely and may not include output device **250**. In other words, device **200** may be “headless” and may not include a display, for example.

[0035] Communication interface **260** may include a transceiver that enables device **200** to communicate with other devices and/or systems via wireless communications (e.g., radio frequency, infrared, and/or visual optics, etc.), wired communications (e.g., conductive wire, twisted pair cable, coaxial cable, transmission line, fiber optic cable, and/or waveguide, etc.), or a combination of wireless and wired communications. Communication interface **260** may include a transmitter that converts baseband signals to radio frequency (RF) signals and/or a receiver that converts RF signals to baseband signals. Communication interface **260** may be coupled to an antenna for transmitting and receiving RF signals.

[0036] Communication interface **260** may include a logical component that includes input and/or output ports, input and/or output systems, and/or other input and output components that facilitate the transmission of data to other devices. For example, communication interface **260** may include a network interface card (e.g., Ethernet card) for wired communications and/or a wireless network interface (e.g., a WiFi) card for wireless communications. Communication interface **260** may also include a universal serial bus (USB) port for communications over a cable, a Bluetooth™ wireless interface, a radio-frequency identification (RFID) interface, a near-field communications (NFC) wireless interface, and/or any other type of interface that converts data from one form to another form.

[0037] As will be described in detail below, device **200** may perform certain operations relating to MIMO rank indicator override. Device **200** may perform these operations in response to processor **220** executing software instructions contained in a computer-readable medium, such as memory **230**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may be implemented within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **230** from another computer-readable medium or from another device. The software instructions contained in memory **230** may cause processor **220** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of, or in combination with, software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0038] Although FIG. **2** shows exemplary components of device **200**, in other implementations, device **200** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. **2**. Additionally, or alternatively, one or more components of device **200** may perform one or more tasks described as being performed by one or more other components of device **200**.

[0039] FIG. **3** is a diagram illustrating exemplary components of MIMO rank indicator override system **155**. The components of MIMO rank indicator override system **155** may be implemented, for example, via processor **220** executing instructions from memory **230**. For example, one or more components of MIMO rank indicator override system **155** may correspond to the structure of processor **220** together with instructions in memory **230** for implementing the functionality of the component. Alternatively, or additionally, some or all the components of MIMO rank indicator override system **155** may be implemented via hard-wired circuitry. For example, one or more components of MIMO rank indicator override system **155** may correspond to the structure of some or all of an ASIC, FPGA, and/or another type of integrated circuit. Furthermore, some or all of the components of MIMO rank indicator override system **155** may be implemented in base station **120**. As shown in FIG. **3**, MIMO rank indicator override system **155** may include a UE device interface **310**, a UE device database (DB) **315**, a UE device classifier **320**, core network interfaces **322**, a classification DB **325**, a MIMO rank analyzer **330**, a thresholds DB **335**, a MIMO rank indicator override manager **340**, a MIMO override DB **345**, and a base station interface **350**.

[0040] UE device interface **310** may be configured to communicate with UE device **110**. For example, UE device interface **310** may receive a measurement report, which includes a MIMO rank indicator, from UE device **110**. As another example, UE device interface **310** may receive

information relating to data flow sessions, associated with UE device **110**, from UE device. UE device interface **310** may store information received from UE device **110** in UE device DB **315**. Exemplary information that may be stored in UE device DB **315** is described below with reference to FIG. **4**. Core network interfaces **312** may be configured to communicate with devices in core network **150**. For example, core network interfaces **322** may receive information relating to data flow sessions for UE device **110** from devices in core network **150** managing the data flows (e.g., from a Session Management Function, User Plane Function, Policy Control Function, etc.), and store the received information in UE device DB **315**.

[0041] UE device classifier **320** may classify UE device **110** as a candidate for MIMO rank indicator override or as a non-candidate for MIMO rank indicator override. As an example, MIMO rank indicator override may be important for particular situations, such as when UE device **110** requires high throughput and/or low latency, and/or when UE device **110** is at particular locations where high throughput and/or low latency are more important. UE device classifier **320** may determine a location for UE device **110**, an application ID for an application session associated with UE device **110**, a QoS parameter for a data flow associated with UE device **110**, a network slice to which UE device **110** has been admitted, a MEC service being used by UE device **110**, and/or other types of information that may be used to classify UE device **110**. UE device classifier **320** may compare the obtained information associated with UE device **110** to information stored in classification DB **325** to determine whether UE device **110** should be classified as a candidate for MIMO rank indicator override.

[0042] As an example, candidate criteria, stored in classification DB **325**, for a MIMO rank indicator override may include a list of specific application types, QoS parameters, network slices, locations, and/or MEC services. If the collected information for UE device **110** includes at least one item on the list, UE device **110** may be classified as a candidate for MIMO rank indicator override. In other implementations, each item on the list may be assigned a weight and all the weights may be combined (e.g., as a straight sum, as a weighted average, etc.) and compared to a classifying threshold. If the items from the collected information for UE device **110** combine to a value higher than the classifying threshold, UE device **110** may be classified as a candidate for a MIMO rank indicator override. If UE device **110** does not satisfy the requirements for a MIMO rank indicator override, UE device **110** may not be classified as a candidate for a MIMO rank indicator override. In order to conserve resources of RAN **130** and/or core network **150** and improve efficiency of MIMO communication, only candidate UE devices **110** may be considered when determining whether to apply a MIMO rank indicator override.

[0043] MIMO rank analyzer **330** may determine, when analyzing a received MIMO rank indicator, whether UE device **110** associated with the received MIMO rank indicator is a candidate for MIMO rank indicator override by checking classification DB **325**. If UE device **110** corresponds to a candidate for MIMO rank indicator override, MIMO rank analyzer **330** may analyze the MIMO rank indicator to determine whether the MIMO rank reported by the MIMO rank indicator is lower than an expected MIMO rank. MIMO rank analyzer **330** may determine an expected MIMO rank for UE device **110** based on at least one of an SRS or a PMI associated with UE device **110**. For example, MIMO rank analyzer **330** may use SRS transmissions received from UE device **110** to determine the uplink signal quality associated with different signal paths between UE device **110** and base station **120**. MIMO rank analyzer **330** may then determine the most appropriate MIMO settings for sending signals to UE device **110** based on the determined uplink signal quality (e.g., since different reference signals are sent using different antenna elements, etc.).

[0044] The PMI may be selected and sent to base station **120** by UE device **110** based on the precoding preferred by UE device **110** and may be based on the MIMO configuration preferred by UE device **110**. Thus, base station **120** may compare the PMI to the MIMO configuration associated with the PMI to determine an expected MIMO rank associated with UE device **110**. MIMO rank analyzer **330** may determine that the expected MIMO rank is higher than the MIMO

rank reported in the MIMO rank indicator and, in response, determine that a MIMO rank indicator override determination process should be carried out. If the expected MIMO rank is not higher than the reported MIMO rank, overriding the MIMO rank indicator for UE device **110** may not be performed.

[0045] MIMO rank analyzer **330** may further check whether requirements for performing a MIMO rank indicator override are satisfied by determining a signal quality value for UE device **110** and/or by determining a cell loading condition associated with UE device **110**. For example, MIMO rank analyzer **330** may determine a channel quality value for UE device **110**, such as, for example, a Channel Quality Indicator (CQI) value, a Reference Signal Received Power (RSRP) value, a Reference Signal Received Quality (RSRQ) value, a Received Signal Strength Indicator (RSSI) value, a Signal to Interference and Noise Ratio (SINR) value, and/or another type of signal quality value. MIMO rank analyzer **330** may compare the determined channel quality value to a channel quality threshold, such as, for example, a CQI threshold, an RSRP threshold, an RSRQ threshold, an RSSI threshold, a SINR threshold, and/or another type of channel quality threshold stored in thresholds DB **335**.

[0046] Furthermore, MIMO rank indicator override system **155** may determine a cell loading condition value associated with UE device **110**, such as, for example, a Physical Resource Block (PRB) utilization rate value, a Transmission Time Interval (TTI) utilization rate value, cell downlink throughput value, cell uplink throughput value, and/or another type of cell loading condition value. MIMO rank analyzer **330** may compare the determined cell loading condition value to a cell loading condition threshold, such as, for example, a PRB utilization rate threshold, a TTI utilization rate threshold, cell downlink throughput threshold, cell uplink throughput threshold, and/or another type of cell loading condition threshold stored in thresholds DB **335**.

[0047] If UE device **110** corresponds to a candidate for MIMO rank indicator override, if the MIMO rank reported in the MIMO rank indicator is lower than an expected MIMO rank, and if UE device **110** satisfies the channel quality and/or cell loading requirement for a MIMO rank indicator override, MIMO rank analyzer **330** may instruct MIMO rank indicator override manager **340** to override the MIMO rank indicator for UE device **110**.

[0048] MIMO rank indicator override manager **340** may instruct base station **120** to override the MIMO rank indicator for UE device **110** using base station interface **350**. Thus, base station interface **350** may interface with base station **120** to receive information from base station **120** to MIMO rank indicator override system **155**, and/or to send instructions to base station **120**. In response to receiving the instruction, base station **120** may select a MIMO rank that is greater than the received MIMO rank indicator for UE device **110** and may include information identifying the selected MIMO rank in DCI sent to UE device **110** at the next scheduling opportunity. MIMO rank indicator override manager **340** may store an indication in MIMO override DB **345** that UE device **110** is associated with a MIMO rank indicator override. Thus, MIMO override DB **345** may store information identifying all UE devices **110**, serviced by base station **120**, which are associated with a MIMO rank indicator override.

[0049] Furthermore, MIMO rank indicator override manager **340** may monitor errors associated with the codewords received by UE device **110** from base station **120** to determine whether the error rate has increased as a result of the higher MIMO rank. For example, MIMO rank indicator override manager **340** may determine whether the BLER associated with at least one codeword is below a BLER threshold. If at least one codeword has a BLER that is below a BLER threshold, MIMO rank indicator override manager **340** may continue to override the MIMO rank indicator for UE device **110**. If no codewords have a BLER that is below a BLER threshold, MIMO rank indicator override manager **340** may determine whether at least one codeword is above a BLER threshold for a set duration (e.g., for at least an x number of time slots or y milliseconds). If at least one codeword is above the BLER threshold for the set duration, MIMO rank indicator override manager **340** may cease overriding the MIMO rank indicator for UE device **110**. Otherwise, MIMO

rank indicator override manager **340** may continue to override the MIMO rank indicator for UE device **110**.

[0050] Although FIG. **3** shows exemplary components of MIMO rank indicator override system **155**, in other implementations, MIMO rank indicator override system **155** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. **3**. Additionally, or alternatively, one or more components of MIMO rank indicator override system **155** may perform one or more tasks described as being performed by one or more other components of MIMO rank indicator override system **155**.

[0051] FIG. **4** illustrates exemplary components of user device DB **315**. As shown in FIG. **4**, user device DB **315** may include one or more UE device records **400**. Each UE device record **400** may store information relating to a particular UE device **110** serviced by base station **120**. UE device record **400** may include a UE device identifier (ID) field **410** and one or more time period records **420**.

[0052] UE device ID field **410** may store information identifying a particular UE device **110**. For example, UE device ID field **410** may store a Mobile Directory Number (MDN), an International Mobile Subscriber Identity (IMSI), a Mobile Station International Subscriber Directory Number (MSISDN), an International Mobile Equipment Identity (IMEI), an ID associated with the subscription for UE device **110** (e.g., a subscription ID, an account number, etc.), and/or another type of ID for the particular UE device **110**.

[0053] Each time period record **420** may store information associated with a particular time period. Time period record **420** may include a time period field **422**, a location field **424**, a QoS field **430**, a network slice field **432**, a MEC service field **434**, an application field **436**, and a MIMO rank field **440**. Time period field **422** may identify a time period during which information relating to UE device **110** was received. Location field **424** may store information identifying a location of UE device **110** during the time period.

[0054] QoS field **430** may store one or more QoS parameters associated with UE device **110** during the time period. For example, QoS parameters field **430** may store a 5G QoS ID (5QI) value for a data flow associated with UE device **110** during the time period. Furthermore, QoS parameters field **430** may store values for one or more additional QoS parameters for the data flow such as, for example, an Allocation and Retention Priority (ARP) value that defines a relative importance of data flow in light of resource limitations; a Reflective QoS Attribute (RQA) value that indicates whether QoS of downlink traffic to UE device **110** is reflected on uplink traffic; a Guaranteed Flow Bit Rate (GFBR) value; a Notification Control value that indicates whether notifications are requested from the RAN when the GFBR cannot be guaranteed; one or more Aggregate Maximum Bit Rate (AMBR) values, such as a per session AMBR, a per UE AMBR, etc.; a Maximum Packet Loss Rate (MPLR) value; and/or other types of QoS parameter values.

[0055] Network slice field **432** may store information identifying a network slice used by UE device **110** during the time period. MEC service field **434** may store information identifying whether UE device **110** used a MEC service during the time period and information identifying the MEC service, if UE device **110** used a MEC service during the time period. Application field **436** may store information identifying whether UE device **110** used an application during the time period and information identifying the application, if UE device **110** used the application during the time period. MIMO rank field **440** may store a MIMO rank identified by a MIMO rank indicator received from UE device **110** during the time period.

[0056] Although FIG. **4** shows exemplary components of user device DB **315**, in other implementations, user device DB **315** may include fewer components, different components, additional components, or differently arranged components than depicted in FIG. **4**.

[0057] FIG. **5** illustrates a flowchart of a process **500** for classifying a UE device as a candidate for MIMO rank indicator override. In some implementations, process **500** of FIG. **5** may be performed by MIMO rank indicator override system **155**. In other implementations, some or all of process **500**

may be performed by another device or a group of devices separate from MIMO rank indicator override system **155**.

[0058] As shown in FIG. 5, process **500** may include selecting a UE device (block **510**). For example, MIMO rank indicator override system **155** may select UE device **110** serviced by base station **120** (e.g., from a list of UE devices **110** attached to, and/or registered with, base station **120**). Process **500** may further include determining an application type associated with the selected UE device (block **520**), determining a QoS parameter associated with the selected UE device (block **530**), determining a network slice associated with the selected UE device (block **540**), and determining a location associated with the selected UE device (block **550**).

[0059] For example, MIMO rank indicator override system **155** may obtain information relating to data streams in a Protocol Data Unit (PDU) session associated with UE device **110**. As an example, MIMO rank indicator override system **155** may access a scheduler at base station **110** to determine QoS parameters associated with data flows for UE device **110**. Furthermore, MIMO rank indicator override system **155** may perform deep packet inspection at base station **120** on packets associated with UE device **110** to determine application IDs associated with data flows for UE device **110**. MIMO rank indicator override system **155** may determine a network slice associated with UE device **110** based on a request for a network slice selection ID (e.g., Network Slice Selection Assistance Information (NSSAI) or a Single-NSSAI, etc.) received from UE device **110** and/or a network slice assignment sent to UE device **110** from core network **150**. Furthermore, MIMO rank indicator override system **155** may determine a location associated with UE device **110** based on a area tracking update sent to base station **120** from UE device **110**.

[0060] Additionally, or alternatively, MIMO rank indicator override system **155** may obtain information relating to QoS parameters, application IDs, and/or network slices associated with data flows for UE device **110** from core network **150** (e.g., from a Session Management Function, User Plane Function, Policy Control Function, etc.).

[0061] MIMO rank indicator override system **155** may determine application types for application IDs associated with UE device **110** based on a list of application IDs maintained by MIMO rank indicator override system **155**. An application type may correspond, for example, to a real-time video application, a streaming application, a mission critical application, a real-time gaming application, etc.

[0062] MIMO rank indicator override system **155** may further obtain additional information relating to UE device **110**. For example, MIMO rank indicator override system **155** may determine whether UE device **110** is using a MEC service associated with MEC network **140** and use the MEC service information to classify UE device **110**. MIMO rank indicator override system **155** may determine whether UE device **110** is associated with a MEC service based on forwarding information obtained from RAN **130** and/or core network **150**.

[0063] Process **500** may further include classifying the selected UE device as a candidate for MIMO rank indicator override based on the determined information (block **560**). For example, MIMO rank indicator override system **155** may determine whether UE device **110** is associated with a high throughput requirement, a low latency requirement and/or another type of requirement that may benefit from a MIMO rank indicator override. MIMO rank indicator override system **155** may use the collected information relating to the location, application type, QoS parameters, network slices, and/or MEC services to classify UE device **110** into a candidate or non-candidate for a MIMO rank indicator override based on a set of one or more criteria stored in classification DB **325**.

[0064] FIG. 6 illustrates a flowchart of a process **600** for performing a MIMO rank indicator override. In some implementations, process **600** of FIG. 6 may be performed by MIMO rank indicator override system **155**. In other implementations, some or all of process **600** may be performed by another device or a group of devices separate from MIMO rank indicator override system **155**.

[0065] As shown in FIG. 6, process **600** may include receiving a MIMO rank indicator from a UE device (block **610**). For example, base station **120** may receive a MIMO rank indicator in a measurement report sent by UE device **110** and provide the received MIMO rank indicator to MIMO rank indicator override system **155**. Process **600** may further include determining that the received MIMO rank indicator is lower than an expected MIMO rank for the UE device (block **620**). For example, MIMO rank indicator override system **155** may determine an expected MIMO rank for UE device **110** based on at least one of an SRS or a PMI associated with UE device **110**. MIMO rank indicator override system **155** may determine that the expected MIMO rank is higher than the MIMO rank reported in the MIMO rank indicator and continue to block **625** to determine that a MIMO rank indicator override determination process should be carried out. In situations in which the expected MIMO rank is not higher than the reported MIMO rank, overriding the MIMO rank indicator for UE device **110** may not be performed and the rest of process **600** may not need to be performed.

[0066] A determination may be made as to whether the UE device is on a MIMO rank indicator override list (block **625**). For example, MIMO rank indicator override system **155** may determine whether UE device **110** is on a candidate list for MIMO rank indicator override maintained in classification DB **325**. If it is determined that the UE device is not on the MIMO rank indicator override list (block **625**—NO), overriding the MIMO rank indicator for the UE device may not be performed (block **630**).

[0067] If it is determined that the UE device is on the MIMO rank indicator override list (block **625**—YES), the channel quality for the UE device may be determined (block **640**), and the cell loading condition for the UE device may be determined (block **650**). For example, MIMO rank indicator override system **155** may determine a channel quality value for UE device **110**, such as, for example, a CQI value, an RSRP value, an RSRQ value, an RSSI value, a SINR value, etc., and compare the determined channel quality value to a channel quality threshold, such as, for example, a CQI threshold, an RSRP threshold, an RSRQ threshold, an RSSI threshold, a SINR threshold, etc. Furthermore, MIMO rank indicator override system **155** may determine a cell loading condition value associated with UE device **110**, such as, for example, a PRB utilization rate value, a TTI utilization rate value, cell downlink throughput value, cell uplink throughput value, etc., and compare the determined cell loading condition value to a cell loading condition threshold, such as, for example, a PRB utilization rate threshold, a TTI utilization rate threshold, cell downlink throughput threshold, cell uplink throughput threshold, etc.

[0068] A determination may be made as to whether the UE device meets MIMO rank indicator override criteria (block **655**). For example, MIMO rank indicator override system **155** may determine whether one or more channel quality threshold requirements and/or one or more cell loading condition threshold are satisfied for UE device **110**. If it is determined that the UE device does not meet the MIMO rank indicator override criteria (block **655**—NO), overriding the MIMO rank indicator for the UE device may not be performed (block **630**).

[0069] If it is determined that the UE device does meet the MIMO rank indicator override criteria (block **655**—YES), overriding the MIMO rank indicator for the UE device may be performed (block **660**). For example, MIMO rank indicator override system **155** may instruct base station **120** to override the MIMO rank indicator for UE device **110**. In response, base station **120** may select a MIMO rank that is greater than the received MIMO rank indicator for UE device **110** and may include information identifying the selected MIMO rank in DCI sent to UE device **110** at the next scheduling opportunity.

[0070] A determination may be made as to whether the BLER for at least one codeword is below a BLER threshold (block **665**). For example, after sending the DCI to UE device **110** with the selected MIMO rank, MIMO rank indicator override system **155** may determine whether the selecting a MIMO rank higher than the reported MIMO rank in the MIMO rank indicator was appropriate by determining whether the reliability of the communication path from base station **120**

to UE device **110** has decreased. MIMO rank indicator override system **155** may determine whether the reliability has decreased by measuring the BLER of the codewords received by UE device **110**. [0071] If it is determined that the BLER for at least one codeword is below a BLER threshold (block **665**—YES), processing may return to block **660** to continue to override the MIMO rank indicator for the UE device. If it is determined that the BLER for at least one codeword is not below a BLER threshold (block **665**—NO), a determination may be made as to whether at least one codeword is above a BLER threshold for a set duration (block **675**). For example, MIMO rank indicator override system **155** may determine whether at least one codeword, of the two codewords associated with the MIMO communication path, is associated with a BLER that is above a BLER threshold for at least an x number of time slots or y milliseconds.

[0072] If it is determined that at least one codeword is above the BLER threshold for a set duration (block **675**—YES), processing may proceed to block **630** to cease overriding the MIMO rank indicator for the UE device. If it is determined that at least one codeword is not above the BLER threshold for a set duration (block **675**—NO), processing may return to block **660** to continue to override the MIMO rank indicator for the UE device.

[0073] FIG. 7 illustrates an exemplary signal flow **700** diagram according to an implementation described herein. As shown in FIG. 7, signal flow **700** includes UE device **110** sending a MIMO rank indicator to base station **120** (signal **710**). Base station **120**, which may include MIMO rank indicator override system **155**, may determine that UE device **110** has not been classified as a candidate or MIMO rank indicator override, because UE device **110** is not associated with high throughput requirement or low latency requirement applications. Thus, base station **120** may select not to override the MIMO rank indicator (block **712**).

[0074] At a later time, UE device **110** may initiate an application session with application server **165** via base station **120** (signals **720** and **722**). The application session may correspond to a video streaming application with a low latency requirement and a high throughput requirement. As a result, base station **120** may classify UE device **110** as a candidate for MIMO rank indicator override. UE device **110** may send another MIMO rank indicator to base station **120** (signal **730**) and base station **120** may detect that UE device **110** is a candidate or MIMO rank indicator override. Base station **120** may check the channel quality for UE device **110**, and/or the loading condition for the cell of base station **120** associated with UE device **110**, and determine that UE device **110** satisfies a channel quality threshold and a cell loading threshold to activate a MIMO rank indicator override. In response, base station **120** may select to override the reported MIMO rank in the received MIMO rank indicator (block **732**) and may override the MIMO rank by including a higher MIMO rank in the DCI information sent to UE device **110** at the next scheduling opportunity (signal **734**).

[0075] Subsequently, base station **120** may measure the channel quality for UE device **110** and may determine that a BLER for a codeword received by UE device **110** is not below a BLER threshold (block **736**). Thus, the higher MIMO rank has resulted in a degraded signal quality. In response, base station **120** may cancel the MIMO rank indicator override by reverting to the lower MIMO rank received in the most recent MIMO rank indicator by including the lower MIMO rank in the DCI information sent to UE device **110** at the next scheduling opportunity (signal **740**).

[0076] At a later time, UE device **110** may move to a new location (block **750**). Alternatively, UE device **110** may stay in the same location and the multipath fading conditions in the location may change (e.g., as a result of traffic, weather, etc.). UE device **110** may send another MIMO rank indicator to base station **120** (signal **760**) and base station **120** may detect that UE device **110** is a candidate or MIMO rank indicator override. Base station **120** may check the channel quality for UE device **110**, and/or the loading condition for the cell of base station **120** associated with UE device **110**, and determine that UE device **110** satisfies a channel quality threshold and a cell loading threshold to activate a MIMO rank indicator override. In response, base station **120** may select to override the reported MIMO rank in the received MIMO rank indicator (block **762**) and may

override the MIMO rank by including a higher MIMO rank in the DCI information sent to UE device **110** at the next scheduling opportunity (signal **764**).

[0077] Subsequently, base station **120** may measure the channel quality for UE device **110** and may determine that a BLER for a codeword received by UE device **110** is below a BLER threshold and that the BLER requirements are satisfied (block **766**). In response, base station **120** may maintain the MIMO rank indicator override.

[0078] In the preceding specification, various preferred embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

[0079] For example, while a series of blocks have been described with respect to FIGS. **5** and **6**, and a series of signals have been described with respect to FIG. **7**, the order of the blocks, and/or signals, may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel.

[0080] It will be apparent that systems and/or methods, as described above, may be implemented in many different forms of software, firmware, and hardware in the implementations illustrated in the figures. The actual software code or specialized control hardware used to implement these systems and methods is not limiting of the embodiments. Thus, the operation and behavior of the systems and methods were described without reference to the specific software code—it being understood that software and control hardware can be designed to implement the systems and methods based on the description herein.

[0081] Further, certain portions, described above, may be implemented as a component that performs one or more functions. A component, as used herein, may include hardware, such as a processor, an ASIC, or a FPGA, or a combination of hardware and software (e.g., a processor executing software).

[0082] It should be emphasized that the terms “comprises”/“comprising” when used in this specification are taken to specify the presence of stated features, integers, steps or components but does not preclude the presence or addition of one or more other features, integers, steps, components or groups thereof.

[0083] The term “logic,” as used herein, may refer to a combination of one or more processors configured to execute instructions stored in one or more memory devices, may refer to hardwired circuitry, and/or may refer to a combination thereof. Furthermore, a logic may be included in a single device or may be distributed across multiple, and possibly remote, devices.

[0084] For the purposes of describing and defining the present invention, it is additionally noted that the term “substantially” is utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. The term “substantially” is also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

[0085] To the extent the aforementioned embodiments collect, store, or employ personal information of individuals, it should be understood that such information shall be collected, stored, and used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage and use of such information may be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

[0086] No element, act, or instruction used in the present application should be construed as critical or essential to the embodiments unless explicitly described as such. Also, as used herein, the article “a” is intended to include one or more items. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A method comprising: receiving, by a device, a multiple-input and multiple-output (MIMO) rank indicator from a user equipment (UE) device; determining, by the device, that the received MIMO rank indicator is lower than an expected MIMO rank indicator; determining, by the device, that the UE device satisfies one or more criteria for a MIMO rank indicator override; and overriding, by the device, the MIMO rank indicator for the UE device, in response to determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator and determining that the UE device satisfies the one or more criteria for the MIMO rank indicator override.
2. The method of claim 1, wherein receiving the MIMO rank indicator from the UE device includes: retrieving the MIMO rank indicator from a measurement report received from the UE device.
3. The method of claim 1, wherein determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator includes: determining the expected MIMO rank indicator based on at least one of a Sounding Reference Signal (SRS) associated with the UE device or a Precoding Matrix Indicator (PMI) associated with the UE device.
4. The method of claim 1, wherein determining that the UE device satisfies one or more criteria for the MIMO rank indicator override includes: determining that a channel quality value associated with the UE device is greater than a channel quality threshold.
5. The method of claim 1, wherein determining that the UE device satisfies one or more criteria for the MIMO rank indicator override includes: determining that a cell loading condition associated with the UE device is greater than a cell loading threshold.
6. The method of claim 1, wherein determining that the UE device satisfies one or more criteria for the MIMO rank indicator override includes: determining that the UE device has been classified as a candidate for the MIMO rank indicator override.
7. The method of claim 6, wherein the UE device is classified as a candidate for the MIMO rank indicator override based on at least one of: an application in use by the UE device, a Quality of Service (QoS) parameter for a data flow associated with the UE device, a network slice in use by the UE device, a Multi-Access Edge Computing (MEC) service in use by the UE device, or a location associated with the UE device.
8. The method of claim 1, wherein overriding the MIMO rank indicator for the UE device includes: selecting a MIMO rank greater than the received MIMO rank indicator for the UE device; and including information identifying the selected MIMO rank in downlink channel information (DCI) sent to the UE device.
9. The method of claim 1, further comprising: determining that a block error rate (BLER) associated with the UE device is less than a BLER threshold; and continuing to override the MIMO rank indicator for the UE device, in response to determining that the BLER associated with the UE device is less than the BLER threshold.
10. The method of claim 1, further comprising: determining that a block error rate (BLER) associated with the UE device is greater than a BLER threshold for a particular number of slots or a particular time period; and ceasing to override the MIMO rank indicator for the UE device, in response to determining that the BLER associated with the UE device is greater than the BLER threshold for the particular number of slots or the particular time period.
11. A device comprising: a processor configured to: receive a multiple-input and multiple-output (MIMO) rank indicator from a user equipment (UE) device; determine that the received MIMO

rank indicator is lower than an expected MIMO rank indicator; determine that the UE device satisfies one or more criteria for a MIMO rank indicator override; and override the MIMO rank indicator for the UE device, in response to determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator and determining that the UE device satisfies the one or more criteria for the MIMO rank indicator override.

12. The device of claim 11, wherein, when determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator, the processor is further configured to: determine the expected MIMO rank indicator based on at least one of a Sounding Reference Signal (SRS) associated with the UE device or a Precoding Matrix Indicator (PMI) associated with the UE device.

13. The device of claim 11, wherein, when determining that the UE device satisfies one or more criteria for the MIMO rank indicator override, the processor is further configured to: determine that a channel quality value associated with the UE device is greater than a channel quality threshold.

14. The device of claim 11, wherein, when determining that the UE device satisfies one or more criteria for the MIMO rank indicator override, the processor is further configured to: determine that a cell loading condition associated with the UE device is greater than a cell loading threshold.

15. The device of claim 11, wherein, when determining that the UE device satisfies one or more criteria for the MIMO rank indicator override, the processor is further configured to: determine that the UE device has been classified as a candidate for the MIMO rank indicator override.

16. The device of claim 15, wherein the UE device is classified as a candidate for the MIMO rank indicator override based on at least one of: an application in use by the UE device, a Quality of Service (QoS) parameter for a data flow associated with the UE device, a network slice in use by the UE device, a Multi-Access Edge Computing (MEC) service in use by the UE device, or a location associated with the UE device.

17. The device of claim 11, wherein, when overriding the MIMO rank indicator for the UE device, the processor is further configured to: select a MIMO rank greater than the received MIMO rank indicator for the UE device; and include information identifying the selected MIMO rank in downlink channel information (DCI) sent to the UE device.

18. The device of claim 11, wherein the processor is further configured to: determine that a block error rate (BLER) associated with the UE device is less than a BLER threshold; and continue to override the MIMO rank indicator for the UE device, in response to determining that the BLER associated with the UE device is less than the BLER threshold.

19. The device of claim 11, wherein the processor is further configured to: determine that a block error rate (BLER) associated with the UE device is greater than a BLER threshold for a particular number of slots or a particular time period; and cease to override the MIMO rank indicator for the UE device, in response to determining that the BLER associated with the UE device is greater than the BLER threshold for the particular number of slots or the particular time period.

20. A non-transitory computer-readable memory device storing instructions executable by a processor, the non-transitory computer-readable memory device comprising: one or more instructions to receive a multiple-input and multiple-output (MIMO) rank indicator from a user equipment (UE) device; one or more instructions to determine that the received MIMO rank indicator is lower than an expected MIMO rank indicator; one or more instructions to determine that the UE device satisfies one or more criteria for a MIMO rank indicator override; and one or more instructions to override the MIMO rank indicator for the UE device, in response to determining that the received MIMO rank indicator is lower than the expected MIMO rank indicator and determining that the UE device satisfies the one or more criteria for the MIMO rank indicator override.
